Note:

i. All questions are compulsory.
ii. Answers to Section-I and Section-II should be written in **Two Separate** answer books.
iii. Questions from Section-I attempted in the answer book of Section-II and vice-versa will not be assessed / not be given any credit.
iv. Draw neat and labelled diagrams wherever necessary.
v. Figures to the right indicate full marks.
vi. Answer to every new question must begin on a new page.

SECTION – I

[BOTANY]

Q.1. Select and write the most appropriate answer from the given alternatives (along with its alphabet number) for each sub-question: **[7]**

i. The phenotypic ratio of incomplete dominance is _____.
(A) 1 : 1 (B) 3 : 1
(C) 1 : 2 : 1 (D) 9 : 3 : 3 : 1

ii. Nucleoside is a nucleotide without _____.
(A) sugar (B) nitrogen base
(C) hydrogen bond (D) phosphate group

iii. Which of the following is white button mushroom?
(A) *Agaricus bisporus* (B) *Pleurotus florida*
(C) *Volvariella volvacea* (D) *Candida* sp.

iv. Brown rust of wheat is caused by _____.
(A) viruses (B) bacteria
(C) fungi (D) aphids

v. The reaction centre of PS-II is _____.
(A) Chl – a, 700 (B) Chl – a, 680
(C) Chl – a, 673 (D) Chl – a, 650

vi. The enzymes required for synthesis of ATP are located on _____.
(A) oxysomes (B) cristae
(C) matrix (D) ribosomes

vii. In a food chain, the herbivores are represented by _____.
(A) producers (B) primary consumers
(C) secondary consumers (D) decomposers

Q.2. (A) Answer each question in 'One' sentence only: **(6)[12]**

i. What is leaching?
ii. Define chemoautotrophs.
iii. Name the cell organelle in which Krebs' cycle occurs.
iv. What is 'deforestation'?
v. Give the microbial source of Vit. B_{12}.
vi. What is primary treatment of sewage?

(B) Sketch and label a pollen grain of angiosperms. **(2)**

(C) **Attempt any TWO of the following:** **(4)**

i. Enlist the basic steps involved in recombinant DNA technology.

ii. Give 'two' examples of microbial pesticides with their hosts.

iii. Give the significance of respiration.

iv. Explain the energy pyramid.

Q.3. **(A)** **Attempt any TWO of the following:** **(6)[9]**

i. With the help of a suitable diagram describe the structure of a nucleosome.

ii. Describe the steps of PCR technique.

iii. Describe different steps involved in tissue culture technique.

(B) Sketch and label T.S. of a leaf showing Kranz anatomy. **(3)**

Q.4. With the help of a neat and labelled diagram describe the development of female gametophyte of angiosperms. **[7]**

OR

Explain "the law of independent assortment" with a suitable example.

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SECTION – II

[ZOOLOGY]

Q.5. Select and write the most appropriate answer from the given alternatives (along with its alphabet number) for each sub-question: **[7]**

i. _____ is used for desalination of brackish water.
(A) Reverse osmosis (B) Ion-exchange
(C) Adsorption (D) Electrodialysis

ii. Which of the following is mesodermal in origin?
(A) Retina (B) Enamel of teeth
(C) Heart (D) Liver

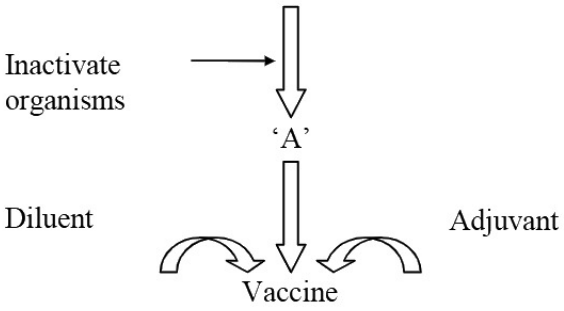
iii. _____ is an exotic breed of cow.
(A) Gir (B) Sindhi
(C) Sahiwal (D) Jersey

iv. _____ maintains basal metabolic rate.
(A) Thyroxine (B) ADH
(C) GH (D) Oxytocin

v. Which of the following is an example of ZW-ZZ type of mechanism of sex determination?
(A) Honey bee (B) Fish
(C) Bird (D) Human being

vi. Transfer of gene between populations that differ genetically from one another is called _____.
(A) Gene mutation (B) Gene flow
(C) Genetic drift (D) Genetic recombination

vii. In the given diagram of vaccine manufacturing process 'A' is _____.



(A) antigen (B) antibody
(C) antitoxin (D) antibiotics

Q.6. (A) Answer each question in 'one' sentence only: **(6)[12]**

i. Longer toes and long prehensile tail indicate which adaptation?
ii. What does 'IUCD' indicate?
iii. Name the valve between left atrium and left ventricle and give its significance.
iv. Give the use of bovine growth hormone.
v. State any 'two' symptoms of Down's syndrome.
vi. Mention any 'one' skeletal difference between ape and man.

(B) Sketch and label the structure of Antibody. **(2)**

(C) Attempt any TWO of the following: **(4)**

i. Name the causative organism of 'typhoid' and draw its diagram.
ii. State the economic importance of 'lac culture'
iii. 'All organisms produce more young ones'. Comment.
iv. Describe 'agranulocytes' with the help of diagrams.

Q.7. (A) Attempt any TWO of the following: **(6)[9]**

i. Define parasitism and give any 'two' types with suitable example of each.
ii. Describe the structure of chromosome with a suitable diagram.
iii. Define 'genomics'. Give any 'two' applications of it.

(B) Sketch and label Malpighian body. **(3)**

Q.8. Describe the histology of 'human testis'. Write a note on human sperm. **[7]**

OR

With the help of a neat and labelled diagram describe the anatomy of 'human eye'. Explain the mechanism of vision.